

Cooperative adaptive cruise control of connected and autonomous vehicles via hybrid iteration

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Abstract—This paper proposes a novel data-driven approach, named hybrid iteration (HI), for the implementation of cooperative adaptive cruise control (CACC) in connected and autonomous vehicles (CAVs). The proposed method leverages the integration of adaptive dynamic programming (ADP), internal model principle, and distributed control technique. By collecting input-state data, an approximately optimal control policy is synthesized through the online iterative technique, HI, to optimize the performance of CACC by minimizing a predefined cost function. Note that the HI preserves the benefits of two established learning algorithms: policy iteration (PI) and value iteration (VI). The learning process initiates with the VI algorithm, followed by the application of the PI algorithm to expedite convergence. HI eliminates the necessity for an initial stabilizing control policy, which is a prerequisite in PI, and exhibits a faster convergence rate than VI. The efficiency and safety of the HI method in CACC systems have been demonstrated through platoon vehicle testing and verification in traffic congestion and cut-in scenarios.

Index Terms—Hybrid iteration, adaptive dynamic programming, connected and autonomous vehicles, cooperative adaptive cruise control.

I. INTRODUCTION

THE progression of autonomous vehicle technology is imperative for the deployment of intelligent transportation systems (ITSs), aiming to reduce traffic accidents made by human drivers and fuel consumption. To further enhance the traffic safety and reliability, the introduction of wireless communication technology into vehicles has sparked significant interest among researchers across disciplines, including control systems [1]–[4], optimization [5]–[7], and communications [8]–[11], in the design of connected and autonomous vehicle (CAV) systems. Among recent research efforts on CAVs to reduce road congestion and improve traffic efficiency, cooperative adaptive cruise control (CACC) is considered as one of the most empirical technologies for ITS applications [12]–[14]. CACC can be regarded as an extension of adaptive cruise control (ACC), enhancing its functionality with vehicle-to-vehicle (V2V) communication and cooperative control. By

leveraging V2V technology, vehicles can collaborate more intelligently to enhance the driving experience as well as the overall efficiency and safety of the traffic system, with CACC relying not only on sensors but also on communication to exchange information, enabling vehicles to adapt to those ahead and mitigate the shockwave effect. A group of vehicles coordinated by CACC is referred to as a vehicular platoon. It is indicated that, in addition to reduced fuel consumption due to smoother traffic flow, a decrease in air resistance can also reduce fuel consumption when vehicles are closer within a platoon [15], [16]. The advent of CAV technologies has greatly improved traffic in terms of energy efficiency and road safety. The ecological control problem of CAV is studied, and the nonlinear optimal control problem is transformed into a convex problem by spatial domain modeling to improve the computational efficiency [17]. By adjusting speed in real-time and maintaining appropriate distances, the overall road efficiency is improved by minimizing sudden braking and acceleration.

There are several control challenges when designing CACC for vehicular platoons, including sensor failures [18], communication loss [19], communication delay [20], [21], and time-varying topologies [22]. To be more specific, in [18], the authors establish a CACC framework that can tolerate sensor failures. By employing switched system theory and Lyapunov stability theory, a robust state-feedback controller is developed and a sufficient condition is presented such that the closed-loop system achieves string stability and zero steady-state error. Considering the case that the intraplatoon communication is lost such that the topology is changed, dynamic communication topology-based CAV-following model is constructed in [19] to maintain the safety of vehicular platoons. In [20], the authors have combined optimal control and CACC to ensure its robustness against communication delay and mobility. In [21], the authors consider a multi-agent Markov decision process with internal delays in CACC, and designs a delay-aware multi-agent reinforcement learning (RL) method for CAVs to enhance platoon safety and stability. In [22], the authors integrate RL with distributed control technique, considering time-varying topological structures, to develop RL-based CACC that ensures stability of the closed-loop system.

Despite the extensive research on CACC, several technical challenges persist its practical implementation, necessitating further investigation. Firstly, current CACC approaches are predominantly model-based. The cornerstone of model-based control policies involves developing a mathematical representation of the system. However, even minor inaccuracies in

This work has been partly supported by the National Key R&D Program of China under Grant No. 2024YFA1012702, the National Natural Science Foundation of China under Grant 62373090 and Grant 61991404, the Research Program of the Liaoning Liaohe Laboratory under Grant LLL23ZZ-05-01, the Key Research and Development Program of Liaoning Province under Grant 2023JH26/10200011, and the Liaoning Revitalization Talents Program under Grant No. XLYC2403177. The corresponding author is Weinan Gao.

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system modeling can lead to significant issues. Controllers derived from inaccurate models may destabilize the vehicle system, posing risks to the safety of autonomous vehicles. Secondly, to enhance the dynamic performance of automated vehicles and to reduce fuel consumption, there is an imperative to design optimal controllers that minimize certain predefined cost functions. A pivotal requirement for most optimal control techniques is precise knowledge of the system model. To overcome the limitation of model-based CACC methods, data-driven CACC has been studied via RL [22]–[24], iterative learning control [25]. RL constitutes a methodology within machine learning that is usually employed for training agents to acquire optimal strategies through learning during their interactions with the environment. RL methods have been used to develop a CACC strategy to safely perform car-following. Function approximation techniques combined with gradient-descent learning algorithms are used as a means to directly modify the control policy and optimize its performance [26]. Policy iteration (PI) and value iteration (VI) are two primary algorithmic approaches within RL that facilitate agents in identifying the optimal behavioral policies to achieve objectives within a specified environment. PI guarantees convergence to the optimal policy after a finite number of iterations, but it requires alternating between policy evaluation and policy improvement, which can result in a significant computational burden. VI is generally more efficient than PI because it updates the value function at each step without the need for an explicit policy representation. However, in some cases, VI may require more iterations to converge to the optimal value function. The hybrid iteration (HI) proposed in this paper effectively combines the advantages of both PI and VI. It first employs VI to learn a stabilizing policy and then initiates PI to accelerate the convergence speed.

Comparing with existing studies on CACC, this paper has three primary contributions.

- Firstly, in contrast to the model-based CACC approaches in the existing literature [27]–[29], we have devised an efficient, data-driven HI learning method for the deployment of CACC in CAVs. The HI algorithm is capable of deriving an optimal controller with a satisfactory transient response even in the absence of explicit vehicle dynamics knowledge, utilizing data gathered online from vehicle trajectories without the requirement for model information.
- Secondly, we have originally combined HI and distributed internal model for CACC design. Internal model principle is an important tool to address output regulation problem that concerns designing a controller such that the closed-loop system rejects external disturbance while asymptotically tracks some predefined references. Considering CAVs modeled as multi-agent systems, distributed control and internal model can be combined to develop distributed internal model. In contrast to centralized control, which is considered vulnerable due to its reliance on data from every autonomous vehicle within the network, distributed internal model-based control policies provide enhanced robustness against communication breakdowns

or information loss. This is because they operate based solely on information from neighboring autonomous vehicles.

- Thirdly, the proposed algorithm outperforms existing RL-based CACC algorithms such as PI and VI. In detail, VI does not require a stabilizing control policy, but its convergence speed is slower than that of PI. PI converges faster than VI, but it necessitates a stabilizing control policy. The proposed control policy in this paper merges the principles of PI and VI. Initially, a stabilizing control policy is inferred through the application of VI, which is subsequently refined by PI to enhance the convergence rate. HI capitalizes on the strengths of both VI and PI, thereby facilitating the realization of the desired control objectives. It can approximate the optimal control policy and corresponding cost function iteratively without needing an accurate knowledge of vehicle dynamics. In this paper, the actual traffic scenarios, such as traffic congestion and vehicle cut-in situations are introduced to deeply verify the efficiency and safety performance of the HI method in CACC. These scenarios are designed to simulate the complex traffic conditions in the real world, so as to comprehensively evaluate the performance of the HI method in dealing with emergency situations and maintaining the safe distance between vehicles. Through validation under complex scenarios, the HI method demonstrates significant performance advantages in CACC systems, effectively enhancing system efficiency while ensuring road safety and driving smoothness.

The rest of this paper is organized as follows. In Section II, we provide the system model and some preliminary knowledge, present the principles of distributed internal models, and review PI and VI algorithms. The design of data-driven CACC and the convergence analysis of algorithm are covered in Section III. In Section IV, simulation and traffic congestion scenarios involving cut-in maneuvers, the effectiveness and superior performance of the proposed control method are verified. Section V contains the conclusion and suggests future research directions.

Notations. Throughout this paper, $\mathbb{R}$ denotes the set of real numbers, while $\mathbb{Z}_+$ denotes the set of nonnegative integers. $\mathcal{P}^n$ denotes the set of all $n \times n$ real, symmetric and positive semidefinite matrices. $\|\cdot\|$ represents the Euclidean norm for vectors and the induced norm for matrices. $\otimes$ indicates the Kronecker product operator and $\text{vec}(A) = [a_1^T, a_2^T, \dots, a_m^T]^T$, where $a_i \in \mathbb{R}^n$ are the columns of $A \in \mathbb{R}^{n \times m}$. For a symmetric matrix $P = P^T \in \mathbb{R}^{m \times m}$, $\text{vecs}(P) = [p_{11}, 2p_{12}, \dots, 2p_{1m}, p_{22}, 2p_{23}, \dots, 2p_{m-1,m}, p_{mm}]^T \in \mathbb{R}^{\frac{1}{2}m(m+1)}$. $\text{vecv}(v) = [v_1^2, v_1v_2, \dots, v_1v_m, v_2^2, v_2v_3, \dots, v_{m-1}v_m, v_m^2]^T \in \mathbb{R}^{\frac{1}{2}m(m+1)}$. $P \succ (\succeq)0$ indicates that P is a positive definite (semidefinite) matrix, $P \prec (\preceq)0$ indicates that P is negative definite (semidefinite).

II. PROBLEM FORMULATION AND PRELIMINARIES

In this paper, we consider a platoon of n vehicles traveling along a straight line. The dynamics of the i -th autonomous

vehicle can be described as follows [30]:

$$\begin{aligned}\dot{p}_i &= v_i, \\ \dot{v}_i &= a_i, \\ \dot{a}_i &= -\tau_i^{-1}a_i + \tau_i^{-1}u_i - \frac{d_{mi}}{m_i}\tau_i^{-1} \\ &:= -\tau_i^{-1}a_i + \tau_i^{-1}u_i + \tilde{d}_i\end{aligned}\quad (1)$$

where p_i represents the position, v_i denotes the speed of vehicle i , a_i is the acceleration, τ_i denotes the time constant associated with its engine, d_{mi} the mechanical drag, and m_i represents the mass of vehicle.

Based on (1), we derive the following state-space representation for the i -th vehicle in the platoon.

$$\dot{x}_i = A_i x_i + B_i u_i + D_i v, \quad i = 1, 2, \dots, N \quad (2)$$

where τ_h is time headway, v^* is desired velocity of the platoon, the state is $x_i^T = [p_i + iv^*\tau_h, v_i, a_i]$, and control input $u_i \in \mathbb{R}$,

$$A_i = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & -\tau_i^{-1} \end{bmatrix}, B_i = \begin{bmatrix} 0 \\ 0 \\ \tau_i^{-1} \end{bmatrix}, D_i = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & \tilde{d}_i \end{bmatrix}.$$

Define e_i is the tracking error and the state equation of the exosystem as follows

$$\dot{e}_i = C_i x_i + Fv, \quad i = 1, 2, \dots, N \quad (3)$$

$$\dot{v} = Ev \quad (4)$$

where

$$C_i = [1 \quad 0 \quad 0], F = [-v^* \quad 0], E = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix},$$

and $e_i \in \mathbb{R}$ are the tracking error of the i -th subsystem. The state vector $v \in \mathbb{R}^2$ represents the state of an exosystem that generates the disturbance $D_i v$ and the reference signal $y_0 = -Fv$, which is to be tracked by the output $y_i = C_i x_i$ of each subsystem. Considering the exosystem (4) and the plant (2), we introduce a directed graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$. The node set $\mathcal{V} = \{0, 1, \dots, N\}$ comprises node 0, which corresponds to the leader governed by the exosystem (4), along with the additional N node, which represent the followers characterized by the plant (2). The edge set $\mathcal{E} \subset \mathcal{V} \times \mathcal{V}$ denotes the connections between the node. The adjacency matrix $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{(N+1) \times (N+1)}$ is defined such that $a_{ij} = 1$ if $(j, i) \in \mathcal{E}$ and otherwise $a_{ij} = 0$. Then, the corresponding Laplacian $\mathcal{L}$ of $\mathcal{G}$ is

$$\mathcal{L} = \begin{bmatrix} \sum_{j=1}^N a_{0j} & -[a_{01} \cdots a_{0N}] \\ -\Delta \mathbf{1}_N & \mathcal{H} \end{bmatrix}$$

where $\Delta = \text{diag}\{a_{10}, a_{20}, \dots, a_{N0}\}$, and $\mathcal{H} = [h_{ij}] \in \mathbb{R}^{N \times N}$ is defined by $h_{ii} = \left(\sum_{j=0}^N a_{ij}\right) - a_{ii}$ and $h_{ij} = -a_{ij}$ for all $i \neq j$. $\mathcal{N}_i$ defines a set of neighbors of the i -th agent, and $\mathcal{N}_i^+ := \mathcal{N}_i \setminus \{0\}$.

The following standard assumptions are required for addressing the CACC problem for CAVs using data-driven approaches.

Assumption 1. (A_i, B_i) is stabilizable, $\forall 1 \leq i \leq N$.

Assumption 2. $\text{rank} \begin{bmatrix} A_i - \lambda I & B_i \\ C_i & 0 \end{bmatrix} = n_i + 1, \forall \lambda \in \sigma(E), \forall 1 \leq i \leq N$.

Assumption 3. The digraph $\mathcal{G}$ contains a directed spanning tree with the node 0 as the root.

If each subsystem independently maintains its own internal model based on data from local sensors and limited external interactions (information from neighboring nodes), the system can operate without relying on global information from a central server or other nodes. This allows for the construction of a decentralized internal model using output regulation theory, as described below:

$$\dot{z}_i = Ez_i + G_2 e_i, \quad i = 1, 2, \dots, N. \quad (5)$$

where $G_2 = [0 \quad 1]^T$.

The implementation of a decentralized internal model (5) is based on the presumption that all followers can directly communicate with the leader to acquire e_i . Nonetheless, this presumption may not hold in light of the communication constraints among agents. Therefore, one can not use e_i as a feedback to design internal model. In our earlier research [31], we define $\hat{e}_i$ as a weighted sum of tracking errors of followers.

$$\hat{e}_i = \sum_{j \in \mathcal{N}_i} \frac{a_{ij}(y_i - y_j)}{\sum_{j=0}^N a_{ij}}. \quad (6)$$

Then, we introduce a distributed internal model as follows

$$\dot{z}_i = Ez_i + G_2 \hat{e}_i, \quad i = 1, 2, \dots, N. \quad (7)$$

The following Lemma 1 is given to show the cooperative output regulation problem [32]–[34] can be addressed by developing a state-feedback controller with distributed internal model.

Lemma 1. [31, Theorem 1] Under Assumptions 1-3, the augmented system

$$\begin{aligned}\dot{v} &= Ev \\ \dot{x}_i &= A_i x_i + B_i u_i + D_i v, \\ \dot{z}_i &= Ez_i + G_2 \hat{e}_i, \\ e_i &= C_i x_i + Fv, \quad i = 1, 2, \dots, N\end{aligned}\quad (8)$$

in closed-loop with the state-feedback controller

$$u_i = -K_{xi} x_i - K_{zi} z_i, \quad i = 1, 2, \dots, N, \quad (9)$$

achieves cooperative output regulation if the matrix

$$A_{ci} = \begin{bmatrix} A_i - B_i K_{xi} & -B_i K_{zi} \\ G_2 C_i & E \end{bmatrix}$$

is Hurwitz.

To facilitate the presentation, we define the following vectors and matrices to consolidate the states, inputs, and system

matrices.

$$\begin{aligned}\tilde{u} &= [\tilde{u}_1, \tilde{u}_2, \dots, \tilde{u}_N]^T, \\ \bar{A} &= \text{blockdiag}(\bar{A}_1, \bar{A}_2, \dots, \bar{A}_N), \\ \bar{B} &= \text{blockdiag}(\bar{B}_1, \bar{B}_2, \dots, \bar{B}_N), \\ Q &= \text{blockdiag}(Q_1, Q_2, \dots, Q_N), \\ R &= \text{blockdiag}(R_1, R_2, \dots, R_N),\end{aligned}$$

where $\tilde{u}_i = u_i - U_i v$, $Q_i = Q_i^T > 0$, $R_i = R_i^T > 0$. Define $U_i = -K_{xi}X_i - K_{zi}Z_i$, for $i = 1, 2, \dots, N$.

$$\begin{aligned}\tilde{x}_i &= x_i - X_i v, \quad \tilde{z}_i = z_i - Z_i v, \\ \tilde{\xi}_i &= \begin{bmatrix} \tilde{x}_i \\ \tilde{z}_i \end{bmatrix} \in \mathbb{R}^5, \quad K_i = \begin{bmatrix} K_{xi} & K_{zi} \end{bmatrix}, \\ \bar{A}_i &= \begin{bmatrix} A_i & 0 \\ G_2 C_i & E \end{bmatrix}, \quad \bar{B}_i = \begin{bmatrix} B_i \\ 0 \end{bmatrix}, i = 1, 2, \dots, N. \quad (10)\end{aligned}$$

Remark 1. Notably, under Assumptions 1-3, the pair $(\bar{A}_i, \bar{B}_i)$ is stabilizable for all $i = 1, 2, \dots, N$. This guarantees the existence of a feedback gain K_i such that the closed-loop matrix A_{ci} is Hurwitz. Furthermore, the matrices (X_i, U_i) correspond to the unique solutions of the regulator equations. For a rigorous proof of this result, see [Lemma 1.26] in [35] for more details.

Now, we can design an optimal controller by minimizing the following cost function.

Problem 1.

$$\min_{\tilde{u}} \int_0^\infty (\tilde{\xi}^T Q \tilde{\xi} + \tilde{u}^T R \tilde{u}) dt \quad (11)$$

$$\text{s.t. } \dot{\tilde{\xi}} = \bar{A} \tilde{\xi} + \bar{B} \tilde{u} \quad (12)$$

where

$$\tilde{\xi} = [\tilde{\xi}_1^T, \tilde{\xi}_2^T, \dots, \tilde{\xi}_N^T]^T, \text{ for } i = 1, 2, \dots, N.$$

Based on the linear optimal control theory, the optimal control policy solving Problem 1 is

$$\tilde{u}^* = -K^* \tilde{\xi}.$$

Since A, B, Q and R are block diagonal matrices, we can derive that K^* is also a block diagonal matrix representing by $K^* = \text{blockdiag}(K_1^*, K_2^*, \dots, K_N^*)$, where the optimal control gain for the i -th follower is

$$K_i^* = R_i^{-1} \bar{B}_i^T P_i^* := \begin{bmatrix} K_{xi}^* & K_{zi}^* \end{bmatrix}. \quad (13)$$

The matrix P_i^* is the unique solution to the following algebraic Riccati equation

$$\bar{A}_i^T P_i^* + P_i^* \bar{A}_i + Q_i - P_i^* \bar{B}_i R_i^{-1} \bar{B}_i^T P_i^* = 0. \quad (14)$$

Therefore, the optimal controller is equivalent to

$$\begin{aligned}u_i^* &= \tilde{u}_i^* + U_i v \\ &= -K_i^* \tilde{\xi}_i + U_i v \\ &= -K_{xi}^* \tilde{x}_i - K_{zi}^* \tilde{z}_i + U_i v \\ &= -K_{xi}^* x_i - K_{zi}^* z_i, \quad i = 1, 2, \dots, N, \quad (15)\end{aligned}$$

which is similar to the controller (9).

Moreover, since the pair $(\bar{A}_i, \bar{B}_i)$ is stabilizable, it is straightforward to have the state matrix of closed-loop system

$$A_{ci}^* = \begin{bmatrix} A_i - B_i K_{xi}^* & -B_i K_{zi}^* \\ G_2 C_i & E \end{bmatrix}$$

is Hurwitz for all the followers. In light of Lemma 1, the optimal controller (15) ensures the cooperative output regulation is achieved. It is worth noting that (14) nonlinear in P_i^* . This makes it difficult to find P_i^* directly from equation (14). To solve (14), two successive approximation methods, VI and PI, can be applied and their details are given in the following subsections.

A. Policy Iteration

The PI algorithm is a commonly used method for approximating the optimal feedback control law. It consists of two steps: policy evaluation and policy improvement. For policy evaluation, Kleinman [36] first introduced the iterative method in linear systems, followed by the use of value functions to estimate the feedback law. For policy improvement, the estimate of the value function is updated to reduce the approximation error. The main ideas of PI algorithm are briefly reviewed below.

- Policy Evaluation: Solve $P_i^{(k)}$ from

$$\begin{aligned}0 &= P_i^{(k)} (\bar{A}_i - \bar{B}_i K_i^{(k)}) + (\bar{A}_i - \bar{B}_i K_i^{(k)})^T P_i^{(k)} + Q_i \\ &\quad + (K_i^{(k)})^T R_i K_i^{(k)} \quad (16)\end{aligned}$$

- Policy Improvement: Update the control gain matrix by

$$K_i^{(k+1)} = R_i^{-1} \bar{B}_i^T P_i^{(k)} \quad (17)$$

B. Value Iteration

Unlike PI, VI eases the learning process by addressing the challenge of knowing the stable control gain matrix $K_i^{(0)}$ for each subsystem. The fundamental approach involves solving the problem by a step-by-step approach. It starts with an initial value matrix $P_i^{(0)}$. The solution is then gradually updated according to value update until the solution meets certain stopping criterion. In this way, one can gradually approach to the optimal value. The iteration process for VI is as follows.

- Value Update: Given $P_i^{(0)}$, update the value matrix using

$$\begin{aligned}P_i^{(k+1)} &\leftarrow \epsilon_k (P_i^{(k)} \bar{A}_i + \bar{A}_i^T P_i^{(k)} \\ &\quad + Q_i - P_i^{(k)} \bar{B}_i R_i^{-1} \bar{B}_i^T P_i^{(k)}) \\ &\quad + P_i^{(k)}, i = 1, 2, \dots, N, \quad (18)\end{aligned}$$

where ϵ_k is deterministic sequence and satisfied

$$\epsilon_k > 0, \quad \sum_{k=0}^{\infty} \epsilon_k = \infty, \quad \sum_{k=0}^{\infty} \epsilon_k^2 < \infty. \quad (19)$$

The convergence proof of the VI algorithm has been provided in [37].

III. MAIN RESULTS

In this Section, we propose a novel HI approaches for CACC design, which is mainly divided into two states. In the first phase, VI is used to find a stabilizing control policy. once the stabilizing control gain is found, the PI is switched until it converges to the optimal solution.

We start from model-based HI for CACC design, then move to data-driven HI.

A. Model-Based HI for CACC

1) Phase 1: Finding a stabilizing control policy.

Throughout the first phase, the value matrix is iteratively updated until a stabilizing control policy is found. It is worth emphasizing that the idea at this phase uses stochastic approximation for value updating. Firstly, $\{\mathfrak{B}_r\}_0^\infty$ is defined as a set non-empty internally bounded set, which satisfies

$$\mathfrak{B}_r \subset \mathfrak{B}_{r+1} \in \mathcal{P}^n, r \in \mathbb{Z}_+, \lim_{q \rightarrow \infty} \mathfrak{B}_r = \mathcal{P}^n.$$

and $\varepsilon_i > 0$ is a small threshold. Additionally, the selected deterministic sequence ϵ_k satisfies condition (19).

Since the primary objective of Phase 1 is to find a stabilizing policy for the i -th subsystem, we can assume that there exists a $\hat{Q}_i$ such that $\hat{Q}_i > Q_i$. Replace $\hat{Q}_i$ with Q_i and continue the update steps until a stabilizing control policy is achieved. Once a stabilizing control policy is obtained, conclude Phase 1 and then implement this stabilizing control policy during Phase 2.

2) Phase 2: The second phase explores the optimal control policy CACC.

In the second phase, PI will be used, using the control policy obtained from the first phase. Repeat the policy evaluation in (16) and the policy improvement in (17) until the value matrix $P_i^{(k)}$ is close enough to $P_i^{(*)}$, $i = 1, 2, \dots, N$. Algorithm 1 gives the detailed steps of the model-based HI algorithm for CACC.

Remark 2. The HI algorithm offers two distinct advantages over traditional algorithms: it exhibits faster learning speed compared to the VI algorithm, and unlike the PI algorithm, it does not require a stabilizing policy to be pre-established.

Remark 3. Algorithm 1 is designed to iteratively optimize the matrices $P_i^{(*)}$ and $K_i^{(*)}$ for developing stable controllers. The outer loop starts at $i = 1$ and continues until $i = N + 1$, ensuring all models are processed. The inner structure comprises two nested loops. The first inner loop, parameters are initialized by selecting a sufficiently small $\varepsilon_i > 0$ and positive-definite matrices $P_i^{(0)}$ and $Q_i^{(0)}$. The matrix P_i is then updated iteratively via the Riccati equation. After each update, $\bar{P}_i^{(k+1)}$ is checked to verify if it remains within the stability region $\mathfrak{B}_r$. If violated, P_i is reset to its initial value $P_i^{(0)}$, and the counter r is incremented. The loop terminates at Step 9 when the convergence criterion is met. The second inner loop focuses on validating the controller gain $K_i^{(*)}$. At Step 12, P_i is further updated using the Lyapunov equation to enforce closed-loop stability. The loop stops at Step 13 when the variation $\|P_i^{(k)} - P_i^{(k-1)}\| < \varepsilon_i$, confirming the

Algorithm 1 Model-Based HI

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1:  $i \leftarrow 1$ 
2: repeat
3:   Select a small  $\varepsilon_i > 0$ ,  $P_i^{(0)} = P_i^{T(0)} \succ 0$ , and  $\hat{Q}_i^{(0)} = \hat{Q}_i^{T(0)} \succ Q_i$ .  $k, r \leftarrow 0$ .
4:   repeat
5:      $\bar{P}_i^{(k+1)} \leftarrow \epsilon_k (P_i^{(k)} \bar{A}_i + \bar{A}_i^T P_i^{(k)} + \hat{Q}_i - P_i^{(k)} \bar{B}_i R_i^{-1} \bar{B}_i^T P_i^{(k)}) + P_i^{(k)}$ ,
6:     if  $\bar{P}_i^{(k+1)} \notin \mathfrak{B}_r$  then  $P_i^{(k+1)} \leftarrow P_i^{(0)}$ ,  $r \leftarrow r + 1$ .
7:     else  $P_i^{(k+1)} \leftarrow \bar{P}_i^{(k+1)}$ , endif
8:      $k \leftarrow k + 1$ 
9:   until  $\bar{P}_i^{(k)} - P_i^{(k-1)} / \epsilon_{k-1} < \hat{Q}_i$ 
10:  repeat
11:     $K_i^{(k)} \leftarrow R_i^{-1} \bar{B}_i^T P_i^{(k-1)}$ 
12:    Solve  $P_i^{(k)}$  from  $P_i^{(k)} (\bar{A}_i - \bar{B}_i K_i^{(k)}) + (\bar{A}_i - \bar{B}_i K_i^{(k)})^T P_i^{(k)} + Q_i + (K_i^{(k)})^T R_i K_i^{(k)} = 0$ .  $k \leftarrow k + 1$ .
13:  until  $\|P_i^{(k)} - P_i^{(k-1)}\| < \varepsilon_i$ 
14:   $i \leftarrow i + 1$ 
15: until  $i = N + 1$ 

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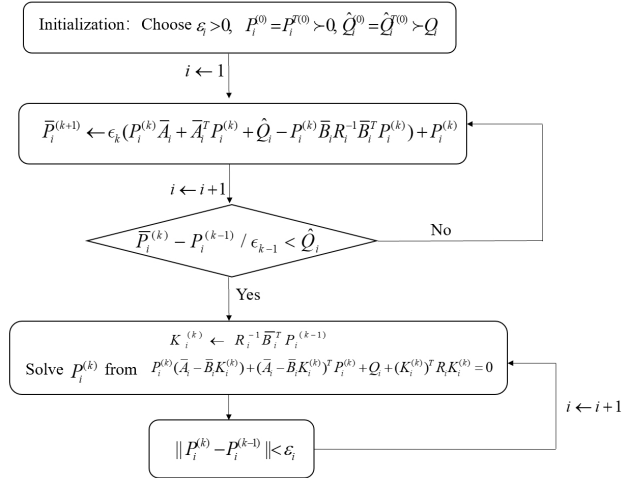


Fig. 1. Illustration of a HI method

convergence of the control gain $K_i^{(*)}$. The algorithm concludes after processing all models.

Theorem 1. Sequences $\{P_i^{(k)}\}_{k=0}^\infty$ and $\{K_i^{(k)}\}_{k=1}^\infty$ computed by Algorithm 1 converge to $P_i^{(*)}$ and $K_i^{(*)}$, $i = 1, 2, \dots, N$, respectively.

Proof: According to [37], As k goes to infinity, we have $\|P_i^{(k)} - P_i^{(*)}\| \rightarrow 0$. Based on the condition in step 9, that is, $\bar{P}_i^{(k)} - P_i^{(k-1)} / \epsilon_{k-1} < \hat{Q}_i$, we can obtain

$$P_i^{(k-1)} \bar{A}_i + \bar{A}_i^T P_i^{(k-1)} - P_i^{(k-1)} \bar{B}_i R_i^{-1} \bar{B}_i^T P_i^{(k-1)} \prec 0,$$

which implies the following inequality:

$$P_i^{(k-1)} (\bar{A}_i - \bar{B}_i K_i^{(k)}) + (\bar{A}_i - \bar{B}_i K_i^{(k)})^T P_i^{(k-1)} \prec (K_i^{(k)})^T R_i K_i^{(k)} \leq 0$$

where $K_i^{(k)} = R_i^{-1} \bar{B}_i^T P_i^{(k-1)}$. Therefore, given any $\hat{Q}_i =$

$\hat{Q}_i^T > 0$, a stabilizing control gain $K_i^{(k_a)} = R_i^{-1} \bar{B}_i^T P_i^{(k_a-1)}$ can always be obtained such that the matrix $\bar{A}_i - \bar{B}_i K_i^{(k_a)}$ is Hurwitz when steps 3-9 are finished at iteration $k_a \in \mathbb{Z}_+$. Starting from the stabilizing control gain $K_i^{(k_a)}$, according to the [36, Theorem], repeating steps 10-13 will lead to the convergence to the optimal solution (P_i^*, K_i^*) . ■

Theorem 2. Sequences $\{P_i^{(k)}\}_{k=0}^\infty$ and $\{K_i^{(k)}\}_{k=1}^\infty$, computed by HI Algorithm 1, converge to P_i^* and K_i^* , respectively, at a quadratic convergence rate for $i = 1, 2, \dots, N$.

Proof: Using Algorithm 1, when switching to PI at steps 10-13, the following results can be obtained from equations (16) and (17) of the PI method. We can get

$$\begin{aligned} & (\bar{A}_i - \bar{B}_i K_i^{(k+1)})^T (P_i^{(k+1)} - P_i^*) \\ & + (P_i^{(k+1)} - P_i^*) (\bar{A}_i - \bar{B}_i K_i^{(k+1)}) \\ & = (\bar{A}_i)^T P_i^{(k+1)} - (\bar{A}_i)^T P_i^* - (K_i^{(k+1)})^T \bar{B}_i^T P_i^{(k+1)} \\ & + (K_i^{(k+1)})^T \bar{B}_i^T P_i^* + P_i^{(k+1)} \bar{A}_i - P_i^{(k+1)} \bar{B}_i K_i^{(k+1)} \\ & - P_i^* \bar{A}_i + P_i^* \bar{B}_i K_i^{(k+1)} + (K_i^*)^T \bar{B}_i^T P_i^* \\ & - (K_i^*)^T \bar{B}_i^T P_i^* + P_i^* \bar{B}_i K_i^* - P_i^* \bar{B}_i K_i^* \\ & = -Q_i - (K_i^{(k+1)})^T R_i K_i^{(k+1)} - (\bar{A}_i - \bar{B}_i K_i^*)^T P_i^* \\ & - P_i^* (\bar{A}_i - \bar{B}_i K_i^*) - (K_i^* - K_i^{(k+1)})^T \bar{B}_i^T P_i^* \\ & - P_i^* \bar{B}_i (K_i^* - K_i^{(k+1)}) \\ & = -Q_i - (K_i^{(k+1)})^T R_i K_i^{(k+1)} + Q_i + K_i^* R_i K_i^* \\ & - (K_i^* - K_i^{(k+1)})^T R_i K_i^* - K_i^* R_i (K_i^* - K_i^{(k+1)}) \\ & = - (K_i^{(k+1)} - K_i^*)^T R_i (K_i^{(k+1)} - K_i^*) \\ & = - (P_i^{(k)} - P_i^*) \bar{B}_i R_i^{-1} (\bar{B}_i)^T (P_i^{(k)} - P_i^*). \end{aligned} \quad (20)$$

According to (20), we can obtain

$$\begin{aligned} P_i^{(k+1)} - P_i^* &= \int_0^\infty e^{(\bar{A}_i - \bar{B}_i K_i^{(k+1)})\tau} (P_i^{(k)} - P_i^*) \\ &\quad \times \bar{B}_i R_i^{-1} (\bar{B}_i)^T (P_i^{(k)} - P_i^*) \\ &\quad - P_i^*) e^{(\bar{A}_i - \bar{B}_i K_i^{(k+1)})\tau} d\tau \end{aligned}$$

This means that there exist constants γ_p and γ_k , such as,

$$\begin{aligned} \lim_{k \rightarrow \infty} \frac{\|P_i^{(k+1)} - P_i^*\|}{\|P_i^{(k)} - P_i^*\|^2} &= \gamma_p, \\ \lim_{k \rightarrow \infty} \frac{\|K_i^{(k+1)} - K_i^*\|}{\|K_i^{(k)} - K_i^*\|^2} &= \gamma_k. \end{aligned}$$

Thus, the sequences $\{P_i^{(k)}\}_{k=0}^\infty$ and $\{K_i^{(k)}\}_{k=1}^\infty$ both converge quadratically to their optimal values. So the proof is done. ■

Remark 4. In the generalized policy iteration (GPI) method, policy evaluation is replaced by iterative approximation; if a sufficiently large number of iterations are used for approximation, it is guaranteed that the updated policy is admissible. Compared to existing PI and GPI methods [38], the proposed HI algorithm no longer relies on an admissible control policy to initiate the learning process. The online implementation

of the GPI assumes the dynamics are known. However, this assumption is completely removed in our proposed off-policy HI Algorithm 2.

B. Data-Driven CACC design

The key point to address Problem 1 is to solve the Riccati equation (14), which usually depends on the knowledge of system matrices. In this section, we will design a data-driven HI algorithm to explore the optimal CACC policy. The developed approaches are capable of approximating the control gains K_i^* for each follower without requiring knowledge of the system matrices A_i, B_i and D_i .

The leader's state information is required by all followers. To accommodate this requirement, we develop the following estimator of leader's state v .

$$\dot{\zeta}_i = E \zeta_i + \mu \sum_{j \in \mathcal{N}_i} a_{ij} (\zeta_j - \zeta_i), i = 1, 2, \dots, N, \quad (21)$$

with $\zeta_0 = v$, where $\mu > 0$, as time approaches infinity, the estimation error $\tilde{\zeta}_i := \zeta_i - v$ converges to zero.

Then, we rewrite the i -th subsystem and use the internal model (7):

$$\begin{aligned} \dot{\xi}_i &= \bar{A}_i \xi_i + \bar{B}_i u_i + \begin{bmatrix} D_i \\ \alpha_{i0} G_2 F \end{bmatrix} v - \sum_{j \in \mathcal{N}_i^+} \alpha_{ij} \begin{bmatrix} 0 \\ G_2 \bar{C}_j \end{bmatrix} \xi_j \\ &= \bar{A}_i^{(k)} \xi_i + \bar{B}_i (K_i^{(k)} \xi_i + u_i) + \begin{bmatrix} D_i \\ \alpha_{i0} G_2 F \end{bmatrix} \zeta_i \\ &\quad - \sum_{j \in \mathcal{N}_i^+} \alpha_{ij} \begin{bmatrix} 0 \\ G_2 \bar{C}_j \end{bmatrix} \xi_j + \begin{bmatrix} -D_i \tilde{\zeta}_i \\ -\alpha_{i0} G_2 F \tilde{\zeta}_i \end{bmatrix} \\ &= \bar{A}_i^{(k)} \xi_i + \bar{B}_i (K_i^{(k)} \xi_i + u_i) + \Delta_{i0} \zeta_i \\ &\quad - \sum_{j \in \mathcal{N}_i^+} \Delta_{ij} \xi_j + \begin{bmatrix} -D_i \tilde{\zeta}_i \\ -\alpha_{i0} G_2 F \tilde{\zeta}_i \end{bmatrix} \end{aligned}$$

where, $\bar{A}_i^{(k)} = \bar{A}_i - \bar{B}_i K_i^{(k)}$, $\bar{C}_j = [C_j \ 0] \in \mathbb{R}^{1 \times 5}$, $\Delta_{ij} = \begin{bmatrix} 0 \\ \alpha_{ij} G_2 \bar{C}_j \end{bmatrix}$ and $\Delta_{i0} = \begin{bmatrix} D_i \\ \alpha_{i0} G_2 F \end{bmatrix}$, and $\alpha_{ij} = a_{ij} / (\sum_{j=0}^N a_{ij})$ for $i = 1, 2, \dots, N$.

By changing the value of μ , the value of $\tilde{\zeta}_i$ can be altered, so it can be denoted as

$$\Upsilon_i(t) := \begin{bmatrix} -D_i \tilde{\zeta}_i \\ -\alpha_{i0} G_2 F \tilde{\zeta}_i \end{bmatrix},$$

By tuning μ , it will converge to zero at any desired rate.

The model can then be approximated by

$$\dot{\xi}_i = \bar{A}_i^{(k)} \xi_i + \bar{B}_i (K_i^{(k)} \xi_i + u_i) + \Delta_{i0} \zeta_i - \sum_{j \in \mathcal{N}_i^+} \Delta_{ij} \xi_j. \quad (22)$$

1) Phase 1 : Find a Stabilizing CACC Policy

Firstly, a data-driven VI method is proposed to solve the CACC problem. The main purpose of using this method is to obtain a stabilizing control policy. Set up a Lyapunov function

$V_k(\xi_i) = \xi_i^T P_i^{(k)} \xi_i$, where $k \in \mathbb{Z}_+, i = 1, 2, \dots, N$, next, we use mathematical techniques, we get the following

$$\begin{aligned} \dot{V}_k(\xi_i) &= \dot{\xi}_i^T P_i^{(k)} \xi_i + \xi_i^T P_i^{(k)} \dot{\xi}_i \\ &= \left(\bar{A}_i^{(k)} \xi_i + \bar{B}_i \left(K_i^{(k)} \xi_i + u_i \right) + \Delta_{i0} \zeta_i - \sum_{j \in \mathcal{N}_i^+} \Delta_{ij} \xi_j \right. \\ &\quad \left. + \Upsilon_i \right)^T P_i^{(k)} \xi_i + \xi_i^T P_i^{(k)} \left(\bar{A}_i^{(k)} \xi_i \right. \\ &\quad \left. + \bar{B}_i \left(K_i^{(k)} \xi_i + u_i \right) + \Delta_{i0} \zeta_i - \sum_{j \in \mathcal{N}_i^+} \Delta_{ij} \xi_j + \Upsilon_i \right) \\ &= \xi_i^T \left(\bar{A}_i^T P_i^{(k)} + P_i^{(k)} \bar{A}_i \right) \xi_i + 2u_i^T R_i K_i^{(k+1)} \xi_i \\ &\quad + 2\zeta_i^T \Delta_{i0}^T P_i^{(k)} \xi_i - 2 \sum_{j \in \mathcal{N}_i^+} \xi_j^T \Delta_{ij}^T P_i^{(k)} \xi_i \\ &\quad + 2\Upsilon_i^T P_i^{(k)} \xi_i. \end{aligned} \quad (23)$$

Letting

$$\begin{aligned} \psi_i &= [\xi_{j1}^T, \xi_{j2}^T, \dots, \xi_{j|\mathcal{N}_i^+|}^T]^T \in \mathbb{R}^{p_i}, \\ \Xi_i^{(k)} &= \left[P_i^{(k)} \Delta_{ij1}, P_i^{(k)} \Delta_{ij2}, \dots, P_i^{(k)} \Delta_{ij|\mathcal{N}_i^+|} \right]^T \in \mathbb{R}^{p_i \times 5}. \end{aligned} \quad (24)$$

Given $\Upsilon_i = 0$, by Kronecker product representation, we get

$$\begin{aligned} \xi_i^T \left(\bar{A}_i^T P_i^{(k)} + P_i^{(k)} \bar{A}_i \right) \xi_i &= (\xi_i^T \otimes \xi_i^T) \text{vec} \left(\bar{A}_i^T P_i^{(k)} \right. \\ &\quad \left. + P_i^{(k)} \bar{A}_i \right), \\ u_i^T R_i K_i^{(k+1)} \xi_i &= (\xi_i^T \otimes u_i^T) (I \otimes R_i) \text{vec}(K_i^{(k+1)}), \\ \zeta_i^T \Delta_{i0}^T P_i^{(k)} \xi_i &= (\xi_i^T \otimes \zeta_i^T) \text{vec}(\Delta_{i0}^T P_i^{(k)}), \\ \sum_{j \in \mathcal{N}_i^+} \xi_j^T \Delta_{ij}^T P_i^{(k)} \xi_i &= (\xi_i^T \otimes \psi_i^T) \text{vec}(\Xi_i^{(k)}). \end{aligned} \quad (25)$$

Let us integrate $t_0 - t_1$ on both sides of (23)

$$\begin{aligned} &\xi_i(t_1)^T P_i^{(k)} \xi_i(t_1) - \xi_i(t_0)^T P_i^{(k)} \xi_i(t_0) \\ &= \int_{t_0}^{t_1} (\xi_i^T \otimes \xi_i^T) \text{vec} \left(\bar{A}_i^T P_i^{(k)} + P_i^{(k)} \bar{A}_i \right) d\tau \\ &\quad + \int_{t_0}^{t_1} 2(\xi_i^T \otimes u_i^T) (I \otimes R_i) \text{vec}(K_i^{(k+1)}) d\tau \\ &\quad + \int_{t_0}^{t_1} 2(\xi_i^T \otimes \zeta_i^T) \text{vec}(\Delta_{i0}^T P_i^{(k)}) d\tau \\ &\quad - \int_{t_0}^{t_1} 2(\xi_i^T \otimes \psi_i^T) \text{vec}(\Xi_i^{(k)}) d\tau \end{aligned} \quad (26)$$

for any a, b and a sufficiently large number s , define the following matrix:

$$\begin{aligned} \delta_a &= [\text{vecv}(a(t_1)) - \text{vecv}(a(t_0)), \dots, \\ &\quad \text{vecv}(a(t_s)) - \text{vecv}(a(t_{s-1}))]^T, \\ \Gamma_{a,b} &= \left[\int_{t_0}^{t_1} a \otimes b d\tau, \int_{t_1}^{t_2} a \otimes b d\tau, \dots, \int_{t_{s-1}}^{t_s} a \otimes b d\tau \right]^T, \\ &\text{where } t_0 < t_1 < \dots < t_s. \end{aligned}$$

According to (26), the following equation can be obtained

$$\Psi_{VI,i} \begin{bmatrix} \text{vec}(\bar{A}_i^T P_i^{(k)} + P_i^{(k)} \bar{A}_i) \\ \text{vec}(K_i^{(k+1)}) \\ \text{vec}(\Delta_{i0}^T P_i^{(k)}) \\ \text{vec}(\Xi_i^{(k)}) \end{bmatrix} = \Phi_{VI,i}^{(k)} \quad (27)$$

where

$$\begin{aligned} \Psi_{VI,i} &= [\Gamma_{\xi_i, \xi_i}, 2\Gamma_{\xi_i, u_i} (I \otimes R_i), 2\Gamma_{\xi_i, \zeta_i}, -2\Gamma_{\xi_i, \psi_i}], \\ \Phi_{VI,i}^{(k)} &= \delta_{\xi_i} \text{vecs} \left(P_i^{(k)} \right). \end{aligned}$$

The uniqueness of the solution to equation (27) is assured under certain rank conditions, as demonstrated in the following section. Due to limited space, we omit the proof of the Lemma 2, which aligns with the proof methodologies presented in [39], [40].

Lemma 2. For all $k \in \mathbb{Z}_+$, if there exists a $s^* \in \mathbb{Z}_+$ such that for all $s > s^*$,

$$\text{rank}([\Gamma_{\xi_i \xi_i}, \Gamma_{\xi_i u_i}, \Gamma_{\xi_i \zeta_i}, \Gamma_{\xi_i \psi_i}]) = 30 + 5p_i, \quad (28)$$

then the matrix $\Psi_{VI,i}$ has full column rank for all $k \in \mathbb{Z}_+$.

Remark 5. To satisfy the rank condition in (28), exploration noise is added to the input during the learning phase. Exploration noise is usually a sum of random noise, random sinusoidal signals, or sinusoidal signals of different frequencies.

2) Phase 2: Explore the Optimal CACC Policy.

Given the stabilizing CACC policy, one can initiate the PI learning process. Consider the Lyapunov function $V_k(\xi_i) = \xi_i^T P_i^{(k)} \xi_i$, and the derivative of $V_k(\xi_i)$ and model (22):

$$\begin{aligned} \dot{V}_k(\xi_i) &= \dot{\xi}_i^T P_i^{(k)} \xi_i + \xi_i^T P_i^{(k)} \dot{\xi}_i \\ &= \left(\bar{A}_i^{(k)} \xi_i + \bar{B}_i \left(K_i^{(k)} \xi_i + u_i \right) + \Delta_{i0} \zeta_i \right. \\ &\quad \left. - \sum_{j \in \mathcal{N}_i^+} \Delta_{ij} \xi_j + \Upsilon_i \right)^T P_i^{(k)} \xi_i \\ &\quad + \xi_i^T P_i^{(k)} \left(\bar{A}_i^{(k)} \xi_i + \bar{B}_i \left(K_i^{(k)} \xi_i + u_i \right) \right. \\ &\quad \left. + \Delta_{i0} \zeta_i - \sum_{j \in \mathcal{N}_i^+} \Delta_{ij} \xi_j + \Upsilon_i \right) \\ &= \xi_i^T \left((\bar{A}_i^{(k)})^T P_i^{(k)} + P_i^{(k)} \bar{A}_i^{(k)} \right) \xi_i \\ &\quad + 2(u_i + K_i^{(k)} \xi_i)^T R_i K_i^{(k+1)} \xi_i + 2\zeta_i^T \Delta_{i0}^T P_i^{(k)} \xi_i \\ &\quad - 2 \sum_{j \in \mathcal{N}_i^+} \xi_j^T \Delta_{ij}^T P_i^{(k)} \xi_i + 2\Upsilon_i^T P_i^{(k)} \xi_i. \end{aligned} \quad (29)$$

The equation (29) above can be rewritten by (16),

$$\begin{aligned} \dot{V}_k(\xi_i) &= -\xi_i^T \left(Q_i + (K_i^{(k)})^T R_i K_i^{(k)} \right) \xi_i \\ &\quad + 2(u_i + K_i^{(k)} \xi_i)^T R_i K_i^{(k+1)} \xi_i + 2\zeta_i^T \Delta_{i0}^T P_i^{(k)} \xi_i \end{aligned}$$

$$-2 \sum_{j \in \mathcal{N}_i^+} \xi_j^T \Delta_{ij}^T P_i^{(k)} \xi_i + 2 \Upsilon_i^T P_i^{(k)} \xi_i. \quad (30)$$

Next, by Kronecker product representation, we get

$$\begin{aligned} \xi_i^T \left(Q_i + (K_i^{(k)})^T R_i K_i^{(k)} \right) \xi_i &= (\xi_i^T \otimes \xi_i) \text{vec} \left(Q_i \right. \\ &\quad \left. + (K_i^{(k)})^T R_i K_i^{(k)} \right), \\ (u_i + K_i^{(k)} \xi_i)^T R_i K_i^{(k+1)} \xi_i &= (\xi_i^T \otimes u_i^T) (I \otimes R_i) \text{vec}(K_i^{(k+1)}) \\ &\quad + (\xi_i^T \otimes \xi_i^T) (I \otimes (K_i^{(k)})^T R_i) \\ &\quad \times \text{vec}(K_i^{(k+1)}), \end{aligned}$$

letting $\Upsilon_i = 0$, we obtain

$$\Psi_{PI,i}^{(k)} \begin{bmatrix} \text{vecs}(P_i^{(k)}) \\ \text{vec}(K_i^{(k+1)}) \\ \text{vec}(\Delta_{i0}^T P_i^{(k)}) \\ \text{vec}(\Xi_i^{(k)}) \end{bmatrix} = \Phi_{PI,i}^{(k)} \quad (31)$$

where

$$\begin{aligned} \Psi_{PI,i}^{(k)} &= [\delta_{\xi_i}, -2\Gamma_{\xi_i, \xi_i} (I \otimes (K_i^{(k)})^T R_i) - 2\Gamma_{\xi_i, u_i} (I \otimes R_i), \\ &\quad - 2\Gamma_{\xi_i, \zeta_i}, 2\Gamma_{\xi_i, \psi_i}], \\ \Phi_{PI,i}^{(k)} &= -\Gamma_{\xi_i, \xi_i} \text{vec} \left(Q_i + (K_i^{(k)})^T R_i K_i^{(k)} \right). \end{aligned}$$

Algorithm 2 Data-Driven HI Algorithm

- 1: $i \leftarrow 1$.
- 2: Choose $\varepsilon_i > 0$, $P_i^{(0)} = P_i^{T(0)} \succ 0$, and $\hat{Q}_i = \hat{Q}_i^T \succ Q_i$
- 3: **repeat**
- 4: Use $u_i = -K_i^{(0)} \xi_i + \mathbf{w}_i$ on $[t_0 - t_s]$ with $\mathbf{w}_i$ an exploration noise.
- 5: **repeat**
- 6: Solve $\bar{A}_i^T P_i^{(k)} + P_i^{(k)} \bar{A}_i$ and $K_i^{(k)}$ from (27).
- 7: $\bar{P}_i^{(k+1)} \leftarrow P_i^{(k)} + \epsilon_k (\bar{A}_i^T P_i^{(k)} + P_i^{(k)} \bar{A}_i + \hat{Q}_i - (K_i^{(k+1)})^T R_i K_i^{(k+1)})$
- 8: **if** $\bar{P}_i^{(k+1)} \notin \mathfrak{B}_r$ **then** $P_i^{(k+1)} \leftarrow P_i^{(0)}$, $r \leftarrow r + 1$.
- 9: **else** $P_i^{(k+1)} \leftarrow \bar{P}_i^{(k+1)}$ **end if**
- 10: $k \leftarrow k + 1$
- 11: **until** $(P_i^{(k)} - P_i^{(k-1)})/\epsilon_k < \hat{Q}_i$,
- 12: **repeat**
- 13: Solve $P_i^{(k)}$ and $K_i^{(k+1)}$ from (31). $k \leftarrow k + 1$.
- 14: **until** $\|P_i^{(k)} - P_i^{(k-1)}\| < \hat{\varepsilon}_i$
- 15: $k \leftarrow k^*$, $j \leftarrow 1$
- 16: The learned controlled is (7), (21) and

$$u_i = K_i^{(k+1)} \xi_i$$

- 17: $i \leftarrow i + 1$
- 18: **until** $i = N + 1$

TABLE I
COMPARISON BETWEEN ALGORITHMS

	HI	VI	PI
Stabilizing K_i^0	Not required	Not required	Required
Convergence rate	Quadratic	Sub-linear	Quadratic

Theorem 3. If (28) is satisfied, sequences $\{P_i^{(k)}\}_{k=0}^\infty$ and $\{K_i^{(k)}\}_{k=1}^\infty$ computed by Algorithm 2 converge to P_i^* and K_i^* for $i = 1, 2, \dots, N$, respectively.

Proof: If the rank condition (28) is satisfied, it can be determined that (27) has a unique solution. Then the pair $((\bar{A}_i^T P_i^{(k)} + P_i^{(k)} \bar{A}_i), K_i^{(k+1)})$ satisfies (23) with $K_i^{(k+1)} = R_i^{-1} \bar{B}_i^T P_i^{(k)}$. Therefore, both $\bar{P}_i^{(k+1)}$ and $P_i^{(k+1)}$ solved from steps 5-11 at each iteration are equivalent to those in steps 4-9 of Algorithm 1. One can also observe that the pair $(P_i^{(k)}, K_i^{(k+1)})$ derived from steps 13-16 is equivalent to the pair produced in steps 9-13 of Algorithm 1 [40]. Since the convergence of $\{P_i^{(k)}\}_{k=0}^\infty$ and $\{K_i^{(k)}\}_{k=1}^\infty$ obtained from Algorithm 1 has been proved in Theorem 1, the convergence of these sequences under the data-driven Algorithm 2 is also ensured. The proof is thus completed. ■

Before closing this section, the differences between the HI, VI and PI algorithms is summarized in Table I.

IV. ILLUSTRATIVE EXAMPLES

To validate our theoretical results, we provide an example involving a platoon of four vehicles.

The initial condition of the exosystem is set by $v = [0 \ 1]^T$. The values of τ_i and $\tilde{d}_i$ are illustrated in Tab. II. The topology of the vehicle platoon is depicted in Figure 2, where each vehicle attempts to maintain a safe distance of 30 meters at steady state. From $t = 0s$ to $t = 1s$, we allow the exostate estimation to converge. We collect data from $t = 1s$ to $t = 2s$ for HI learning. The convergence of P_i for each vehicle is shown in Figure 3. By applying the data-driven HI Algorithm 2, we determine that the control policy is admissible when satisfying the condition in step 12. To be more specific, an admissible control policy is achieved at the iteration $k_1 = 376$, $k_2 = 349$, $k_3 = 323$, and $k_4 = 288$, for $i = 1, 2, 3, 4$, respectively.

The comparison of the learned value $P_i^{(k)}$ with its optimal value $P_i^{(*)}$ of each vehicle is shown in the Figure 3 which indicates that all vehicles can learn towards the corresponding optimal control values by iteration. According to the definition of quadratic convergence, we introduce a parameter $r_k = \frac{|P_i^{k+1} - P_i^*|}{|P_i^k - P_i^*|^2}$, from Figure 4, we see that r_k converges to constant 4.3 after 4 iterations, which shows that the proposed method retains the quadratic convergence rate. Figures 5 and 6 demonstrate that the HI algorithm maintains robust performance across different initial conditions. The solid line represents the actual trajectory of the connected autonomous vehicles, whereas the dashed line corresponds to their reference trajectory. The learned control policy is applied after $t = 2s$ and the vehicle trajectories can be checked in Figure 5, showing that all vehicles can follow their references

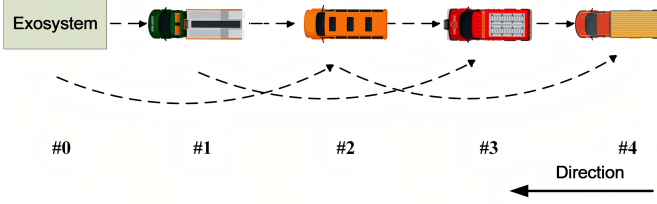


Fig. 2. The structure of a vehicular platoon.

after $t = 12s$, which demonstrates the efficiency of the proposed data-driven control approach. As shown in Figures 7 and 8, under identical test conditions, the data-driven HI method achieves the target speed of 15 m/s within 20 seconds, whereas the method in [41] requires nearly 30 seconds to reach the same velocity. The integral of squared error (ISE) is introduced as a performance metric for quantifying tracking accuracy, where a smaller ISE indicates better system performance in minimizing deviations between actual and desired speeds. Notably, in dynamic scenarios, the proposed data-driven HI-based CACC demonstrates a 4 times reduction in ISE compared to the data-driven CACC method reported in reference [41]. This comparative analysis demonstrates the faster convergence speed and enhanced control efficiency of our data-driven approach in CACC implementation.

To demonstrate the effectiveness and safety of our algorithm, we will incorporate two scenarios with CACC-equipped vehicles.

Firstly, in Figure 9, we observe that the vehicles track their desired speeds during cooperative driving operations. However, when traffic congestion occurred at the 40-second mark, the speed exhibited a significant decrease, demonstrating that the vehicles executed braking maneuvers to adapt to changing traffic conditions. As the congestion gradually dissipated, indicating that the vehicles steadily resumed their original speed.

Secondly, we test the performance of the proposed control method via cut-in scenarios. In Figure 10 verifies the scenario when the two vehicles cut in. At 50 seconds, cut-in vehicle 1 and cut-in vehicle 2 simultaneously merged into the vehicle platoon. To preserve formation integrity, vehicle 3 and vehicle 4 proactively decelerated, ultimately enabling all six vehicles to maintain a collision-free coordinated motion pattern. This validates the effectiveness of the data-driven control method proposed in this paper and demonstrates that it can mitigate the interference caused by the cutting-in vehicle.

TABLE II
SYSTEM PARAMETERS

Parameter	Value	Parameter	Value	Parameter	Value
$\tau_1[s]$	0.1	$\tau_2[s]$	0.11	$\tau_3[s]$	0.12
$\tau_4[s]$	0.13	$\tilde{d}_1[m/s^3]$	3	$\tilde{d}_2[m/s^3]$	3.1
$\tilde{d}_3[m/s^3]$	3.2	$\tilde{d}_4[m/s^3]$	3.3		

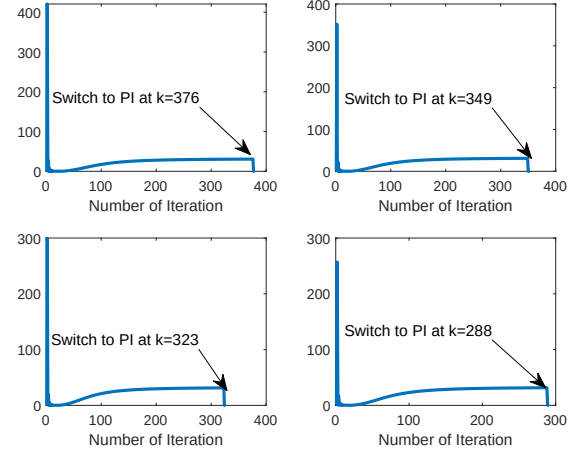


Fig. 3. $\|P_i^{(k)} - P_i^{(*)}\|$ of each follower ($i = 1, 2, 3, 4$) using HI Algorithm.

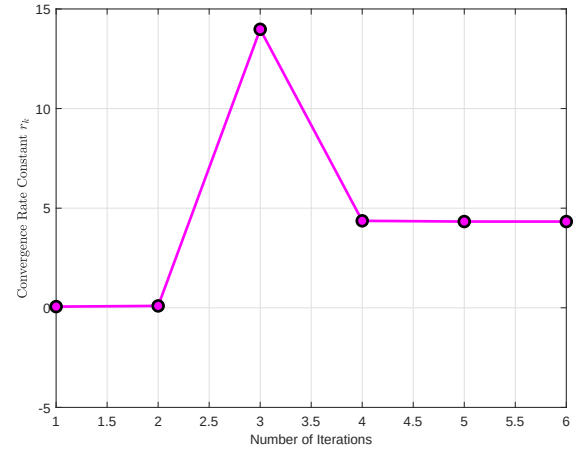


Fig. 4. The evolution of convergence rate constant of the vehicle #4.

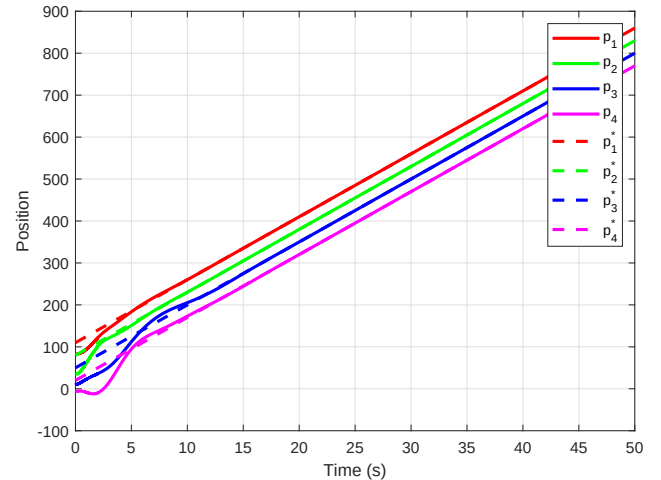


Fig. 5. Trajectories of CAVs and their references with the initial condition $[p_1(0), p_2(0), p_3(0), p_4(0)] = [83, 35, 10, 7]$.

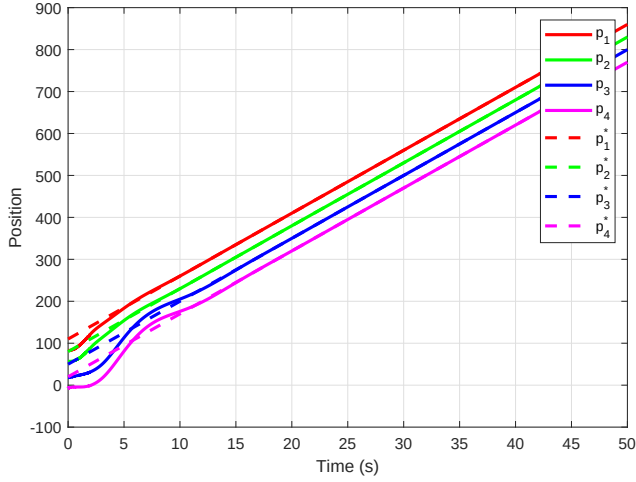


Fig. 6. Trajectories of CAVs and their references with the initial condition $[p_1(0), p_2(0), p_3(0), p_4(0)] = [83, 57, 18, 7]$.

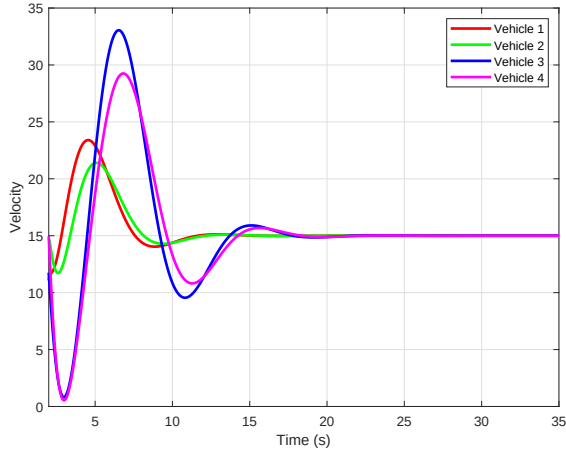


Fig. 7. Velocities of vehicles controlled by Algorithm 2

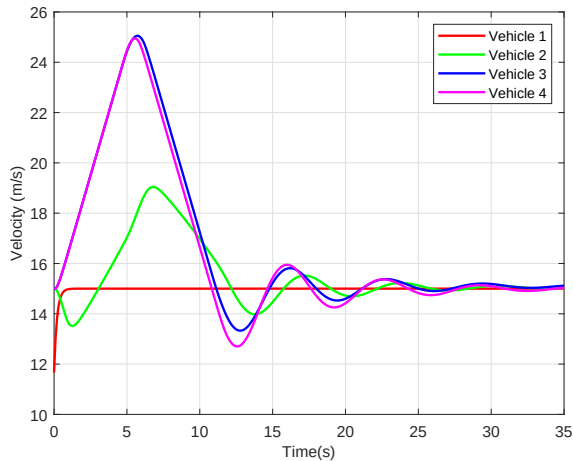


Fig. 8. Velocities of vehicles controlled by data-driven CACC method in [41].

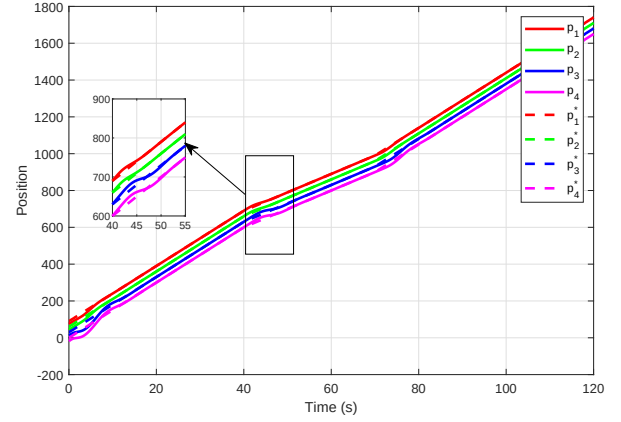


Fig. 9. Trajectories of CAVs during congestion.

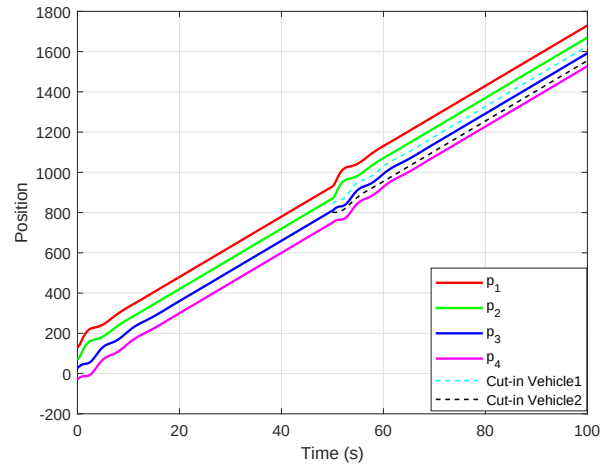


Fig. 10. Trajectory of CAVs in two-vehicle cut-in scenarios.

V. CONCLUSION

This paper proposes a distributed data-driven control method, hybrid iteration, using internal model principle and reinforcement learning techniques to address cooperative adaptive cruise control (CACC) problems for connected and autonomous vehicles, in the absence of precise knowledge of system dynamics, the desired controller is obtained from data through reinforcement learning. Moreover, through the verification under specific scenarios, the high-efficiency and safety performance of the HI method in the CACC have been shown. The limitation of the study is that it involves vehicle simulation and has not been applied in practice. Future work includes the development of advanced CACC methods, the use of hybrid driving and autonomous driving with non-linear models, and their application in real-world environments.

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